



Practice Problem For Linear and Logistic Regression

CSE 457: Machine Learning



Practice Problem For Linear Regression

CSE 457: Machine Learning

Problem Statement



Perform 2 iteration of the gradient descent to get updated weights w_0, w_1, w_2, w_3 .
[Initial $w_0 = w_1 = w_2 = w_3 = 1$ and $\alpha = 0.05$]

x1	x2	x3	y
2	7	6	6.5
4	8	7.5	7.8
6	9	8.5	8.8

Problem Solution

Prediction equation/activation function,

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$

Error function,

$$e = \frac{1}{2} \sum_{i=1}^3 (\hat{y}(i) - y(i))^2$$

General equation for slope for a weight w_j ,

$$s_j = \partial e / \partial w_j = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_j(i)$$

Update weight ,

$$w_j = w_j - \alpha \cdot s_j$$

Where, $w_0 = w_1 = w_2 = w_3 = 1$ and $\alpha = 0.05$

x1	x2	x3	y
2	7	6	6.5
4	8	7.5	7.8
6	9	8.5	8.8

Problem Solution

here,

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$

$$\hat{y} = 1 + x_1 + x_2 + x_3 \quad [w_0 = w_1 = w_2 = w_3 = 1]$$

x1	x2	x3	y
2	7	6	6.5
4	8	7.5	7.8
6	9	8.5	8.8

Problem Solution

here,

$$\hat{y} = 1 + x_1 + x_2 + x_3$$

x1	x2	x3	y
2	7	6	6.5
4	8	7.5	7.8
6	9	8.5	8.8

Problem Solution

here,

$$\hat{y} = 1 + x_1 + x_2 + x_3$$

Slope for a weight w_0 ,

$$s_0 = \partial e / \partial w_0 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_0(i)$$

x1	x2	x3	y
2	7	6	6.5
4	8	7.5	7.8
6	9	8.5	8.8

Problem Solution

here,

$$\hat{y} = 1 + x_1 + x_2 + x_3$$

Slope for a weight w_0 ,

$$s_0 = \partial e / \partial w_0 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_0(i)$$

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	16	9.5
4	8	7.5	7.8	20.5	12.7
6	9	8.5	8.8	23.5	15.7

Problem Solution

here,

$$\hat{y} = 1 + x_1 + x_2 + x_3$$

Slope for a weight w_0 ,

$$\begin{aligned} s_0 &= \partial e / \partial w_0 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_0(i) \\ &= 9.5 \cdot 1 + 12.7 \cdot 1 + 15.7 \cdot 1 \\ s_0 &= 37.9 \end{aligned}$$

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	16	9.5
4	8	7.5	7.8	20.5	12.7
6	9	8.5	8.8	23.5	15.7

Problem Solution

here,

$$\hat{y} = 1 + x_1 + x_2 + x_3$$

Slope for a weight w_0 ,

$$\begin{aligned} s_0 &= \partial e / \partial w_0 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_0(i) \\ &= 9.5 \cdot 1 + 12.7 \cdot 1 + 15.7 \cdot 1 \\ s_0 &= 37.9 \end{aligned}$$

Update weight w_0 ,

$$w_0 = w_0 - \alpha \cdot s_0 = 1 - 0.05 \cdot 37.9 = 1 - 1.895 = -0.895 \approx -0.90$$

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	16	9.5
4	8	7.5	7.8	20.5	12.7
6	9	8.5	8.8	23.5	15.7

Problem Solution

here,

$$\hat{y} = 1 + x_1 + x_2 + x_3$$

Slope for a weight w_1 ,

$$\begin{aligned} s_1 &= \partial e / \partial w_1 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_1(i) \\ &= 9.5 \cdot 2 + 12.7 \cdot 4 + 15.7 \cdot 6 \\ s_1 &= 189.4 \end{aligned}$$

Update weight w_1 ,

$$w_1 = w_1 - \alpha \cdot s_1 = 1 - 0.05 \cdot 189.4 = 1 - 9.47 = -8.47$$

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	16	9.5
4	8	7.5	7.8	20.5	12.7
6	9	8.5	8.8	23.5	15.7

Problem Solution

here,

$$\hat{y} = 1 + x_1 + x_2 + x_3$$

Slope for a weight w_2 ,

$$\begin{aligned} s_2 &= \partial e / \partial w_2 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_2(i) \\ &= 9.5 \cdot 7 + 12.7 \cdot 8 + 15.7 \cdot 9 \\ s_2 &= 309.4 \end{aligned}$$

Update weight w_2 ,

$$w_2 = w_2 - \alpha \cdot s_2 = 1 - 0.05 \cdot 309.4 = 1 - 15.47 = -14.47$$

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	16	9.5
4	8	7.5	7.8	20.5	12.7
6	9	8.5	8.8	23.5	15.7

Problem Solution

here,

$$\hat{y} = 1 + x_1 + x_2 + x_3$$

Slope for a weight w_3 ,

$$\begin{aligned} s_3 &= \partial e / \partial w_3 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_3(i) \\ &= 9.5 \cdot 6 + 12.7 \cdot 7.5 + 15.7 \cdot 8.5 \\ s_3 &= 285.7 \end{aligned}$$

Update weight w_3 ,

$$w_3 = w_3 - \alpha \cdot s_3 = 1 - 0.05 \cdot 285.7 = 1 - 14.285 = -13.285 \approx -13.29$$

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	16	9.5
4	8	7.5	7.8	20.5	12.7
6	9	8.5	8.8	23.5	15.7

Problem Solution

After 1st iteration, weights are,

$$w_0 = -0.90, w_1 = -8.47, w_2 = -14.47, w_3 = -13.29$$

So, updated equation is,

$$\hat{y} = -0.9 - 8.47x_1 - 14.47x_2 - 13.29x_3$$

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	16	9.5
4	8	7.5	7.8	20.5	12.7
6	9	8.5	8.8	23.5	15.7

Problem Solution

After 1st iteration, weights are,

$$w_0 = -0.90, w_1 = -8.47, w_2 = -14.47, w_3 = -13.29$$

So, updated equation is,

$$\hat{y} = -0.9 - 8.47x_1 - 14.47x_2 - 13.29x_3$$

Update table values,

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	-198.87	-205.37
4	8	7.5	7.8	-250.215	-258.015
6	9	8.5	8.8	-294.915	-303.715

Problem Solution

For 2nd iteration,

$$\hat{y} = -0.9 - 8.47x_1 - 14.47x_2 - 13.29x_3$$

Slope for a weight w_0 ,

$$\begin{aligned} s_0 &= \partial e / \partial w_0 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_0(i) \\ &= -205.37 \cdot 1 - 258.015 \cdot 1 - 303.715 \cdot 1 \\ s_0 &= -767.1 \end{aligned}$$

Update weight w_0 ,

$$w_0 = w_0 - \alpha \cdot s_0 = -0.9 - 0.05 \cdot (-767.1) = -0.9 + 38.355 = 37.455 \approx 37.46$$

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	-198.87	-205.37
4	8	7.5	7.8	-250.215	-258.015
6	9	8.5	8.8	-294.915	-303.715

Problem Solution

$$\hat{y} = -0.9 - 8.47x_1 - 14.47x_2 - 13.29x_3$$

Slope for a weight w_1 ,

$$\begin{aligned} s_1 &= \partial e / \partial w_1 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_1(i) \\ &= -205.37 \cdot 2 - 258.015 \cdot 4 - 303.715 \cdot 6 \\ s_1 &= -3265.09 \end{aligned}$$

Update weight w_1 ,

$$w_1 = w_1 - \alpha \cdot s_1 = -8.47 - 0.05 \cdot (-3265.09) = -8.47 + 163.2545 = 154.7845 \approx 154.78$$

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	-198.87	-205.37
4	8	7.5	7.8	-250.215	-258.015
6	9	8.5	8.8	-294.915	-303.715

Problem Solution

$$\hat{y} = -0.9 - 8.47x_1 - 14.47x_2 - 13.29x_3$$

Slope for a weight w_2 ,

$$\begin{aligned} s_2 &= \partial e / \partial w_2 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_2(i) \\ &= -205.37 \cdot 7 - 258.015 \cdot 8 - 303.715 \cdot 9 \\ s_2 &= -6235.145 \end{aligned}$$

Update weight w_2 ,

$$w_2 = w_2 - \alpha \cdot s_2 = -14.47 - 0.05 \cdot (-6235.145) = -14.47 + 311.7573 = 297.2873 \approx 297.29$$

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	-198.87	-205.37
4	8	7.5	7.8	-250.215	-258.015
6	9	8.5	8.8	-294.915	-303.715

Problem Solution

$$\hat{y} = -0.9 - 8.47x_1 - 14.47x_2 - 13.29x_3$$

Slope for a weight w_3 ,

$$\begin{aligned} s_3 &= \partial e / \partial w_3 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_3(i) \\ &= -205.37 \cdot 6 - 258.015 \cdot 7.5 - 303.715 \cdot 8.5 \\ s_3 &= -5748.91 \end{aligned}$$

Update weight w_3 ,

$$w_3 = w_3 - \alpha \cdot s_3 = -13.29 - 0.05 \cdot (-5748.91) = -13.29 + 287.4455 = 274.1555 \approx 274.16$$

x1	x2	x3	y	$\hat{y}$	$\hat{y} - y$
2	7	6	6.5	-198.87	-205.37
4	8	7.5	7.8	-250.215	-258.015
6	9	8.5	8.8	-294.915	-303.715

Problem Solution



After 2nd iteration, weights are,

$$w_0 = 37.46, w_1 = 154.78, w_2 = 297.29, w_3 = 274.16$$

So, updated equation is,

$$\hat{y} = 37.46 + 154.78x_1 + 297.29x_2 + 274.16x_3$$



Practice Problem For Logistic Regression

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Problem Statement



Perform 2 iteration of the gradient descent to get updated weights w_0, w_1, w_2, w_3 .
[Initial $w_0 = w_1 = w_2 = w_3 = 1$ and $\alpha = 0.05$]

x1	x2	x3	y
2	-7	6	No
4	-8	7.5	Yes
6	-9	8.5	Yes

Problem Solution

Activation function (sigmoid function),

$$\hat{y} = 1/(1+e^{-z})$$

Where z is,

$$z = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$

Error function,

$$e = \sum_{i=1}^3 (-y(i)\log(\hat{y}(i)) - (1 - y(i))\log(1 - \hat{y}(i)))$$

General equation for slope for a weight w_j ,

$$s_j = \partial e / \partial w_j = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_j(i)$$

Update weight ,

$$w_j = w_j - \alpha \cdot s_j$$

Where, $w_0 = w_1 = w_2 = w_3 = 1$ and $\alpha = 0.05$

x1	x2	x3	y
2	-7	6	No
4	-8	7.5	Yes
6	-9	8.5	Yes

Problem Solution

here,

$$z = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$

$$z = 1 + x_1 + x_2 + x_3 \quad [w_0 = w_1 = w_2 = w_3 = 1]$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

x1	x2	x3	y
2	-7	6	No (0)
4	-8	7.5	Yes (1)
6	-9	8.5	Yes (1)

Problem Solution

here,

$$z = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$

$$z = 1 + x_1 + x_2 + x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

x1	x2	x3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	2	0.881	0.881
4	-8	7.5	1	4.5	0.989	-0.011
6	-9	8.5	1	6.5	0.998	-0.002

Problem Solution

here,

$$z = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$

$$z = 1 + x_1 + x_2 + x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

Slope for a weight w_0 ,

$$\begin{aligned} s_0 &= \partial e / \partial w_0 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_0(i) \\ &= 0.881 \cdot 1 + (-0.011) \cdot 1 + (-0.002) \cdot 1 \\ s_0 &= 0.868 \end{aligned}$$

Update weight w_0 ,

$$w_0 = w_0 - \alpha \cdot s_0 = 1 - 0.05 \cdot 0.868 = 1 - 0.0434 = 0.9566 \approx 0.96$$

x1	x2	x3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	2	0.881	0.881
4	-8	7.5	1	4.5	0.989	-0.011
6	-9	8.5	1	6.5	0.998	-0.002

Problem Solution

here,

$$z = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$

$$z = 1 + x_1 + x_2 + x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

Slope for a weight w_1 ,

$$\begin{aligned} s_1 &= \partial e / \partial w_1 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_1(i) \\ &= 0.881 \cdot 2 + (-0.011) \cdot 4 + (-0.002) \cdot 6 \\ s_1 &= 1.706 \end{aligned}$$

Update weight w_1 ,

$$w_1 = w_1 - \alpha \cdot s_1 = 1 - 0.05 \cdot 1.706 = 1 - 0.0853 = 0.9147 \approx 0.91$$

x1	x2	x3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	2	0.881	0.881
4	-8	7.5	1	4.5	0.989	-0.011
6	-9	8.5	1	6.5	0.998	-0.002

Problem Solution

here,

$$z = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$

$$z = 1 + x_1 + x_2 + x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

Slope for a weight w_2 ,

$$\begin{aligned} s_2 &= \partial e / \partial w_2 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_2(i) \\ &= 0.881*(-7) + (-0.011)*(-8) + (-0.002)*(-9) \\ s_2 &= -6.061 \end{aligned}$$

Update weight w_2 ,

$$w_2 = w_2 - \alpha \cdot s_2 = 1 - 0.05*(-6.061) = 1 + 0.30305 = 1.3031 \approx 1.30$$

x1	x2	x3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	2	0.881	0.881
4	-8	7.5	1	4.5	0.989	-0.011
6	-9	8.5	1	6.5	0.998	-0.002

Problem Solution

here,

$$z = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$

$$z = 1 + x_1 + x_2 + x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

Slope for a weight w_3 ,

$$\begin{aligned} s_3 &= \partial e / \partial w_3 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_3(i) \\ &= 0.881 \cdot 6 + (-0.011) \cdot 7.5 + (-0.002) \cdot 8.5 \end{aligned}$$

$$s_3 = 5.1865$$

Update weight w_3 ,

$$w_3 = w_3 - \alpha \cdot s_3 = 1 - 0.05 \cdot 5.1865 = 1 - 0.2593 = 0.7407 \approx 0.74$$

x1	x2	x3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	2	0.881	0.881
4	-8	7.5	1	4.5	0.989	-0.011
6	-9	8.5	1	6.5	0.998	-0.002

Problem Solution

After 1st iteration, weights are,

$$w_0 = 0.96, w_1 = 0.91, w_2 = 1.3, w_3 = 0.74$$

Updated activation function,

$$z = 0.96 + 0.91x_1 + 1.3x_2 + 0.74x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

x1	x2	x3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	2	0.881	0.881
4	-8	7.5	1	4.5	0.989	-0.011
6	-9	8.5	1	6.5	0.998	-0.002

Problem Solution

After 1st iteration, weights are,

$$w_0 = 0.96, w_1 = 0.91, w_2 = 1.3, w_3 = 0.74$$

Updated activation function,

$$z = 0.96 + 0.91x_1 + 1.3x_2 + 0.74x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

Updated Table values,

x1	x2	x3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	-1.88	0.132	0.132
4	-8	7.5	1	-0.25	0.438	-0.562
6	-9	8.5	1	1.01	0.733	-0.267

Problem Solution

here,

$$z = 0.96 + 0.91x_1 + 1.3x_2 + 0.74x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

Slope for a weight w_0 ,

$$\begin{aligned} s_0 &= \partial e / \partial w_0 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_0(i) \\ &= 0.132 \cdot 1 + (-0.562) \cdot 1 + (-0.267) \cdot 1 \\ s_0 &= -0.697 \end{aligned}$$

Update weight w_0 ,

$$w_0 = w_0 - \alpha \cdot s_0 = 0.96 - 0.05 \cdot (-0.697) = 0.96 + 0.0349 = 0.9949 \approx 0.99$$

x1	x2	x3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	-1.88	0.132	0.132
4	-8	7.5	1	-0.25	0.438	-0.562
6	-9	8.5	1	1.01	0.733	-0.267

Problem Solution

here,

$$z = 0.96 + 0.91x_1 + 1.3x_2 + 0.74x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

Slope for a weight w_1 ,

$$\begin{aligned} s_1 &= \partial e / \partial w_1 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_1(i) \\ &= 0.132 \cdot 2 + (-0.562) \cdot 4 + (-0.267) \cdot 6 \\ s_1 &= -3.586 \end{aligned}$$

Update weight w_1 ,

$$w_1 = w_1 - \alpha \cdot s_1 = 0.91 - 0.05 \cdot (-3.586) = 0.91 + 0.1793 = 1.0893 \approx 1.09$$

x1	x2	x3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	-1.88	0.132	0.132
4	-8	7.5	1	-0.25	0.438	-0.562
6	-9	8.5	1	1.01	0.733	-0.267

Problem Solution

here,

$$z = 0.96 + 0.91x_1 + 1.3x_2 + 0.74x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

Slope for a weight w_2 ,

$$\begin{aligned} s_2 &= \partial e / \partial w_2 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_2(i) \\ &= 0.132*(-7) + (-0.562)*(-8) + (-0.267)*(-9) \\ s_2 &= 5.975 \end{aligned}$$

Update weight w_2 ,

$$w_2 = w_2 - \alpha s_2 = 1.3 - 0.05*5.975 = 1.3 - 0.2988 = 1.0012 \approx 1.0$$

x1	x2	x3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	-1.88	0.132	0.132
4	-8	7.5	1	-0.25	0.438	-0.562
6	-9	8.5	1	1.01	0.733	-0.267

Problem Solution

here,

$$z = 0.96 + 0.91x_1 + 1.3x_2 + 0.74x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$

Slope for a weight w_3 ,

$$\begin{aligned} s_3 &= \partial e / \partial w_3 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_3(i) \\ &= 0.132 \cdot 6 + (-0.562) \cdot 7.5 + (-0.267) \cdot 8.5 \\ s_3 &= -5.6925 \end{aligned}$$

Update weight w_3 ,

$$w_3 = w_3 - \alpha \cdot s_3 = 0.74 - 0.05 \cdot (-5.6925) = 0.74 + 0.2846 = 1.0246 \approx 1.02$$

x1	x2	x3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	-1.88	0.132	0.132
4	-8	7.5	1	-0.25	0.438	-0.562
6	-9	8.5	1	1.01	0.733	-0.267

Problem Solution



After 2nd iteration, weights are,

$$w_0 = 0.99, w_1 = 1.09, w_2 = 1.0, w_3 = 1.02$$

Updated activation function,

$$z = 0.99 + 1.09x_1 + x_2 + 1.02x_3$$

Sigmoid function,

$$\hat{y} = 1/(1+e^{-z})$$



THE END